

NiCE Text Grid Troubleshooting Guide Version 20.2.0.2

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Sanity Checklist

The following is a list of general things to check to ensure you have a sane Text Grid installation. This is a good place to start when troubleshooting any issue.

- Refresh the Services control panel and verify all the Text Grid services are running.
 - Run `tg status` from a command prompt to verify that an instance of each Text Grid service is found.
 - Use MMC Disk Management to ensure available disk space meets sizing guidelines.
 - Use Event Viewer to verify there are no errors from the Text Grid services in the Windows Logs (both Application and System).
-

Installation Problems

This section covers problems that may occur during installation.

During installation, the following error occurs: Failed to grant 'logon as service' rights to user.

Resolution:

- Ensure the service user name is correct and exists on the server.
 - Ensure a backslash ('\') was used rather than a forward slash (/) in the service user name (e.g., NT AUTHORITY\NetworkService).
-

General Problems

This section covers problems that are not specific to a particular service. If a solution isn't found here, consult the service-specific section(s).

Services will not start.

Symptom(s):

- The service does not start, showing "Error 1069: The service did not start due to a logon error".

Resolution:

- Rerun the configurer, select the correct username and password, and reinstall the services, or
- In the Windows Services application, right click on each service, and select Properties. Adjust the username and password in the Logon Tab.

Symptom(s):

- The service terminates unexpectedly, either when starting manually or as indicated in the System log in the Event Viewer.

Resolution:

- Ensure the configurator app has been run from a command prompt with the correct settings.
- Ensure the user supplied in the configurator has the necessary permissions to write to the <data location> specified during installation.

Nuisance WARN messages show up in service logs.

Symptom(s):

Any of the following messages appear in the service logs:

- No base packages specified - no classpath scanning will be done
- session timeout... is less than connection timeout

Resolution:

- These are benign messages that can be safely ignored.

ERROR or WARN message shows up in service logs: An existing connection was forcibly closed by the remote host

Resolution:

- This indicates the service lost the ability to communicate with another Text Grid service.
- Verify that firewall or network reconfiguration did not change.
- Use the `tg status` command to verify all expected Text Grid services are running.

Index corruption in versions 20.1.0+

Symptom(s):

- An index has a status of red and no available snapshots are available for restoration.

Resolution:

Note In Text Grid versions 20.1.0 and later, indices are split based on the id of the document. Each index contains a maximum of 2^{20} (1,048,576) documents.

- If the index is determined to be corrupted, the first step is to make note of the index name, i.e. `nxtg_chat_en_6_3_336m`
- Delete the index
- Multiply the number in the index name by 2^{20} to determine the id of the first document in the index. In this case, $336 * 2^{20} = 352321536$
- Add 2^{20} to the id of the first document in the index to determine the id of the last document in the index. In this case, $352321536 + 1048576 = 353370112$
- Using NIA, reingest the documents in the range of 352321536 to 353370112. This will recreate the corrupted index, in this case `nxtg_chat_en_6_3_336m`.

Search Service

Search service will not start.

Symptom(s):

- No search logs are produced, and the Windows Event Viewer in the Application log for the NxTgSearch service says **Terminated with unexpected JVM exit code. Check log files for details.**

Resolution:

- Ensure that the machine has enough memory (see System Requirements doc) and that the JVM memory for the search node is appropriately set (as documented in the System Requirements doc)
- In one case, this same error occurred because the service user did not have appropriate AD permissions to access the data directory after an OS upgrade.

Symptom(s):

- The following error appears in the Search `nxtgsearch-stderr*.log` file: **ERROR: will not overwrite keystore at**

[C:\ProgramData\Nexidia\TextGrid2\search\config\elasticsearch.keystore], because this incurs changing the file owner

Resolution:

- Uncheck the "Use Service Account as elasticsearch.keystore owner" checkbox and then "Install Services".

Symptom(s):

- The following error appears in either the Gateway log file or the Windows Event Viewer in the Application log for the NxTgSearch service: **java.lang.IllegalStateException: Failed to obtain node lock, is the following location writable?: [<data location>\search\data\elasticsearch]**

Resolution:

- Ensure the configurator app has been run from a command prompt with the correct settings.
- Ensure the Text Grid service user has the necessary permissions to write to the <data location> specified during installation.

Symptom(s):

- The following error appears in either the Search log file or the Windows Event Viewer in the Application log for the NxTgSearch service: **java.lang.IllegalStateException: The index [...] was created with version [5.4.1] but the minimum compatible version is [6.0.0-beta1]. It should be re-indexed in Elasticsearch 6.x before upgrading to 7.4.0.**

Resolution:

- Ensure that the <datadir> for your Text Grid 12.4 install is different from the <datadir> for any Text Grid 1.1.x install you are upgrading.
- Ensure the migration app has been properly run (see Administration Guide section on migration).

Gateway Service

Gateway service will not start.

Symptom(s):

- Event Viewer contains the following error in the Application log for the NxTgGateway service: **Timed out waiting on managed thread to complete**

Resolution:

- Ensure the configurator app has been run from a command prompt with the correct settings.
- Ensure the Text Grid service user has the necessary permissions to write to the <data location> specified during installation.

Unable to access Gateway REST API

Resolution:

- Ensure the Gateway service is running.
- Ensure network traffic to the Gateway host on the public API port is not being blocked by a firewall.

Initialization problems due to Locks

Background

In order to support multiple Gateways, locks are maintained in Elasticsearch. While the logs may show some lock related error messages, most of these are normal, and will resolve during a normal initialization. However, if the gateway port or machine is changed, or a gateway in a multiple gateway configuration is removed from service, then if that previous gateway instance crashed or was killed at a very specific window of code, then it is possible that all gateway REST requests will fail with a `TgUninitializedException` exception that references this section.

Resolution:

- Verify this is the problem by running http://localhost:9200/_cat/indices. If one of the indices listed is `lockindexmigration` then this is the issue.
- Ensure the Search service is running.
- Disable the Gateway service.
- Run the following Curl command (replacing 127.0.0.1 with the search node host ip) to delete the offending lock.

```
curl -X DELETE 127.0.0.1:9200/lockindexmigration
```

- Note that depending on what security you have installed on the Search node, certain additional parameters may be necessary.

SECURE COMMUNICATION

tg/tg-ui cannot communicate with gateway that has SSL enabled.

Symptom(s):

The Gateway service is configured with SSL enabled and uses the self-signed SSL certificate included with the distribution, the service appears to be running without error but when issuing the `tg status` (or its tg-ui equivalent), the following error comes up:

```
Gateway -----
Nodes:
https://10.1.50.142:28002 - DOWN
I/O error on GET request for "https://10.1.50.142:28002/actuator/health"
: PKIX path building failed: sun.security.provider.certpath.SunCertPathBuilderException: unable to find valid
certification path to requested target; nested exception is javax.net.ssl.SSLHandshakeException: PKIX path
building failed: sun.security.provider.certpath.SunCertPathBuilderException: unable to find valid certification
path to requested target
```

Resolution:

One scenario that can lead to this condition is mis-configuration of intra-Gateway service secure communication. Specifically, when running the Gateway service with the default self-signed SSL certificate, any client of the Gateway service, include the Gateway service itself, needs to bypass the hostname verification step of the HTTPS handshake. Upon user issuing the `tg status` command, the responding Gateway service will get a list of Gateway nodes from the ES store, then query each node's status through the Rest interface on HTTPS.

If getting the above error, check the `server.ssl.lenient-ssl-validation` setting in the Gateway service configuration file `application.yml` and make sure it is set to `true`. Correct any mistake and re-start the service to fix the problem.

TOO_MANY_REQUESTS ERROR DURING MIGRATION

Documents stop migrating due to TOO_MANY_REQUESTS errors during upgrade data migration

Symptoms(s):

During the data migration during an upgrade, the migration process and log show that documents aren't being brought from the Text Grid 1.1 Storage cluster to the Text Grid 2 Search cluster due to a TOO_MANY_REQUESTS errors.

Resolution:

Even though the error itself is vague, TOO_MANY_REQUESTS indicates that the data drive doesn't have enough space to facilitate the migration. This is because the data drive needs to contain the data from the Text Grid 1.1 Storage cluster and the Text Grid 2 Search cluster at the same time. Ensure that the data drive has at least 2 times more space when performing an in-place data migration.

DATA DIFFERENTIAL BETWEEN TG1 AND TG2+ AFTER MIGRATION

There is a document count differential bigger than the threshold after data migration

Symptom(s):

After data migration, there might be a document count differential that is just outside of the threshold determined to be acceptable for the given upgrade, i.e. 1.8% differential where the threshold is 1%.

Resolution:

Documents stored in TG1/TG2+ have states associated with them. Only documents with their state set as INDEXED will be brought over during data migration. The states of missing documents should be checked in this case. Using the example above, of 1.8% of missing documents from TG2+, 1.3% of them have states that aren't INDEXED and were skipped during migration. This means that only .5% of relevant documents are missing from the migration, which is well within the 1% threshold for missing documents.

SEARCH PROBLEMS

Search fails due to offset limit being surpassed

Symptoms(s):

For extremely large documents using highlighting, it's possible to receive a 413 (PAYLOAD_TOO_LARGE) error similar to the following in the response:

```
{
  "timestamp": "2024-05-19 02:26:17.024",
  "code": "payload_too_large",
  "resource_uri": "http://10.1.47.139:28002/search",
  "exception": "TgTooLargeException",
  "message": "The field you are trying to highlight is too large to be analyzed. Please check ntxtgrid.log and set max_analyzed_offset",
  "errors": [
    {
      "details": "The field you are trying to highlight is too large to be analyzed. Please check ntxtgrid.log and set max_analyzed_offset",
      "cause": null,
      "stackPreview": [
        {
          "classLoaderName": "app",
          "moduleName": null,
          "moduleVersion": null,
          "methodName": "buildDocumentSearchImpl",
          "lineNumber": 153,
          "nativeMethod": false,
          "className": "com.nexidia.grid.text.service.impl.document.DocumentSearchImpl"
        },
        {
          "classLoaderName": "app",
          "moduleName": null,
          "moduleVersion": null,
          "methodName": "search",
          "lineNumber": 71,
          "nativeMethod": false,
          "className": "com.nexidia.grid.text.service.impl.document.DocumentSearchImpl"
        }
      ]
    }
  ]
}
```

Check the ntxtgrid.log for a log such as

```
[2024-05-19T02:26:16,859][ERROR][c.n.t.e.h.NxTgHighlighter] [null] Error highlighting field The length [1257291] of field [content] in doc[0]/index[nxtg_general_en_4_3] exceeds the [index.highlight.max_analyzed_offset] limit [1000000]. To avoid this error, set the query parameter [max_analyzed_offset] to a value less than index setting [1000000] and this will tolerate long field values by truncating them.
```

Resolution:

Check the Admin Guide section on [Getting Highlights From Very Large Documents](#).

Search fails due to maxClauseCount being surpassed

Symptom(s):

Textgrid search fails or does not return some expected hits. The error in ntextgrid.log includes "Caused by: org.apache.lucene.search.IndexSearcher\$TooManyNestedClauses: Query contains too many nested clauses; maxClauseCount is set to"

Analysis:

As of TG 12.4.3 (based on Elasticsearch 8.11), Elasticsearch changed the algorithm that protects itself from overly-complicated queries. This means that some queries that may have passed before are now being blocked.

Per <https://github.com/elastic/elasticsearch/issues/88281> the former ability to directly set the max_clause_count has been replaced with computing this value based on a formula of RAM size / thread pool size.

It is not entirely clear what makes a query exceed this limit, but lots of OR clauses and wildcards both seem to have an impact.

Resolution:

Best resolution would be to simplify the query. However, if that is not possible, then either increase the RAM size or decrease the search thread pool count.

The thread pool being referred to is the search pool, which per <https://www.elastic.co/guide/en/elasticsearch/reference/8.9/modules-threadpool.html> is computed as $\text{int}((\# \text{ of allocated processors} * 3) / 2) + 1$.

Search thread pool size can be modified by setting this in the elasticsearch.yaml file (where 10 below is an appropriate lower value than the formula above):

```
thread_pool:
  search:
    size: 10
```

Ram size can be configured using the treeconfigurer or editing as below:

Edit the <datadir>\TextGrid2\search\config\jvm.options.d\heap.options file on each search node and change the current values of

```
-Xms30g
-Xmx30g
```

to

```
-Xms48g
-Xmx48g
```

and then restart the NICE Text Grid 2 Search services.

Note ES recommends keeping the memory below 32G and less than half of the size of the memory in the machine. For the first customer where this occurred, we went to 48G as these were 96G servers and we had not yet figured out how to modify the thread pool size.

Note The command line parameters to treeconfigurer do not allow setting higher than 30G. However, it is still possible to use treeconfigurer to set higher values from the UI.